Biswaranjan Aich

Principal Software Engineer , Dell EMC Bangalore

Bengaluru, India | aich.biswa@gmail.com | +91-9920593782

Professional Summary

Principal Software Engineer with 12+ years of experience architecting large-scale, distributed systems and cloud-native microservices in banking and object storage domains. Designed and led the migration of monolithic applications into 15+ microservices, improving deployment frequency by 60% and reducing system latency by 35%. Hands-on expertise in Java (8 & 11), Golang, Spring Boot, Kubernetes, Docker, and AWS, with deep knowledge of scalable API design, event-driven architecture (Kafka), and secure authentication using OAuth2 and JWT. Serve as an **individual contributor and technical mentor to a 10-member engineering team, member of Architectural team** and conducting architecture reviews, enforcing coding standards, and guiding delivery across multiple agile squads. Built CI/CD pipelines that accelerated release cycles and implemented DevOps practices to enhance reliability and compliance. Currently pursuing an e-Masters in AI & ML (Generative AI specialization) to integrate next-gen intelligence into enterprise-scale systems.

Core Competencies

- Cloud Technologies: AWS (EC2, S3, IAM, VPC, Route 53, SNS, Load Balancers, Auto Scaling)
- Languages: Java 8 and 11, Golang
- Backend: Spring Boot, Spring Cloud, Spring Security, Microservices, REST APIs designing, Multithreading, Kafka
- Frontend: React.js
- DevOps and Release Automation: Docker, Kubernetes, Spinnaker, Jenkins, GitHub Copilot, Codium AI, Checkmarx, Black Duck, PMD, JaCoCo.
- Databases: MySQL, Oracle, MongoDB, Cassandra, Redis
- Testing & Automation: JUnit, Mockito, Power Mock, Robot Framework.
- Architecture Patterns: Circuit Breakers, Service Discovery, Sidecar, Event-driven, Distributed Caching
- Methodologies: Agile/Scrum, BDD, CI/CD
- Security: OAuth2.0, JWT, Keycloak
- Monitoring & Observability: AppDynamics, Prometheus, Grafana, Splunk, AWS CloudWatch.

Professional Experience

GEO Replication & BGP-Load Balancer

DELL EMC, Bangalore
Principal Software Engineer | May 2022 – Present

• Developing high-performance, concurrent applications in Java 8, leveraging multithreading and Stream API for efficient data processing in object storage systems, reducing latency during data retrieval and improving the throughput of read/write operations.
• Utilizing Java 8 and 11 features like Lambda expressions, Streams, and Optional to simplify and optimize complex data manipulation tasks within the object storage domain, making the codebase more concise, readable, and efficient.
• Building and consuming RESTful Web Services to facilitate communication between microservices and external systems, ensuring seamless data flow and integration.
• Enhanced geo-replication for object and metadata writes across multi-site object storage clusters, ensuring eventual consistency and durability while supporting multi-TB daily data ingestion with sub-second write acknowledgements.
• Built and optimized asynchronous replication pipelines between geographically distributed clusters, handling object data, metadata, and versioning information, reducing cross-region replication lag by ~30–40% through batching and parallel replication strategies.
• Developed failure detection and recovery mechanisms for geo-replication workflows, including retry logic, checkpointing, and reconciliation jobs, ensuring minimum data loss (RPO ≈ 10 sec) and faster recovery during network partitions or node failures.
• Collaborated on cluster-level consistency, quorum handling, and metadata synchronization, validating replication correctness under node failures, rolling upgrades, and high-concurrency workloads, improving overall system resilience and operational stability.
• Authored technical specifications aligned with business objectives, drove system design improvements, and mentored junior engineers through code reviews and technical knowledge-sharing sessions to elevate engineering standards, code quality, and team capability.
• Designed and implemented a high-performance BGP-driven Anycast load balancer in Go, optimizing route advertisements and peer-health signaling to improve traffic distribution and reduce failover time by ~35%.
• Developed a Go ↔ Java gRPC interoperability framework leveraging HTTP/2 and Protocol Buffers to enable strongly typed, low-latency communication, reducing API latency by 40–50% compared to REST/JSON.
• Designed performance profiling suites using Prometheus, and Grafana to identify bottlenecks in BGP update paths and gRPC serialization, delivering a ~20% reduction in p99 tail latency through targeted optimizations.
• Designing and implementing multi-container Pods within Kubernetes for handling complex workloads in object storage systems, creating Kubernetes manifest files to define the necessary configurations for deployments, services, and persistent volumes, and managing Kube-config to ensure secure and seamless access to Kubernetes clusters for development and deployment activities.

• Integrating AWS S3 for scalable object storage and implementing SNS for event-driven communication, enabling seamless notification and processing of object storage operations (e.g. file uploads, deletions) across distributed microservices.
• Configuring Auto Scaling Groups and Load Balancers (ALB) to automatically scale microservices based on traffic.
• Implementing load balancing strategies using Kubernetes native features like Services (Cluster IP, Load Balancer, Node Port), Ingress controllers, and NGINX, optimizing traffic distribution for object storage applications.
• Integrating security features within the microservices architecture using Spring Security and OAuth2 to ensure secure authentication and authorization of users and services.
• Designing and implementing Kafka producers and consumers in microservices to handle event-based triggers (such as metadata updates) ensuring reliable message delivery and decoupling services for improved scalability and fault tolerance.

Dynamix (DMX)
Wells Fargo, Bangalore
Software Engineer | Aug 2020 – May 2022

- Orchestrated the integration of microservices using the Kafka messaging system, ensuring reliable communication and data streaming across distributed services.
- Led the conversion of a monolithic application to a microservices architecture, successfully creating, integrating, and testing 10+ microservices to enhance system scalability, flexibility, and maintainability.
- Configured Kafka producers and consumers, overseeing the setup and maintenance of message queues to streamline data flow and improve microservices communication.
- Designed and implemented scalable and secure AWS cloud infrastructure by provisioning and managing services like VPC, EC2, S3, RDS, Route 53, Load Balancer, SNS, and CloudWatch; ensured high availability, proactive monitoring, and disaster recovery to support resilient and efficient operations.
- Configured and deployed Content Delivery Network (Akamai) for faster and more efficient content delivery, enhancing user experience across regions.
- Utilized Redis to cache real-time user trading portfolio snapshots, ensuring low-latency access for high-frequency trading operations and reducing the load on core trading services.

Education

- Master of Computer Applications (MCA) – VSSUT, Burla, Odisha, 2012
- e-Masters in Artificial Intelligence and Machine Learning – IIIT Bangalore (Ongoing)

Technical Leadership

- Operated in a dual role as a hands-on Individual Contributor and Technical Lead, actively designing, coding, and optimizing critical components of large-scale, distributed, cloud-native systems.
- Served as a core member of the architecture team, driving system design, technology selection, and microservices best practices across multiple agile squads. Managed task assignments and cross-functional team coordination.
- Led and mentored a 10+ member engineering team through architecture reviews, code reviews, and technical decision-making, ensuring high-quality, scalable, and secure solutions.
- Collaborated closely with product owners, DevOps, and cross-functional onsite stakeholders to translate business requirements into technical roadmaps and successful deliveries.
- Established engineering standards, CI/CD best practices, and DevOps guidelines, improving release velocity, system reliability, and overall engineering efficiency.

Career Timeline

- May 2022 – Present: DELL EMC, Bengaluru – Principal Software Engineer
- Aug 2020 – May 2022: Wells Fargo, Bengaluru – Software Engineer
- Apr 2018 – Aug 2020: Barclays, Mumbai – Senior Software Developer
- Nov 2016 – Apr 2018: Bank of America, Mumbai – Senior Software Engineer
- Mar 2013 – Nov 2016: Capgemini, Mumbai – Technical Consultant